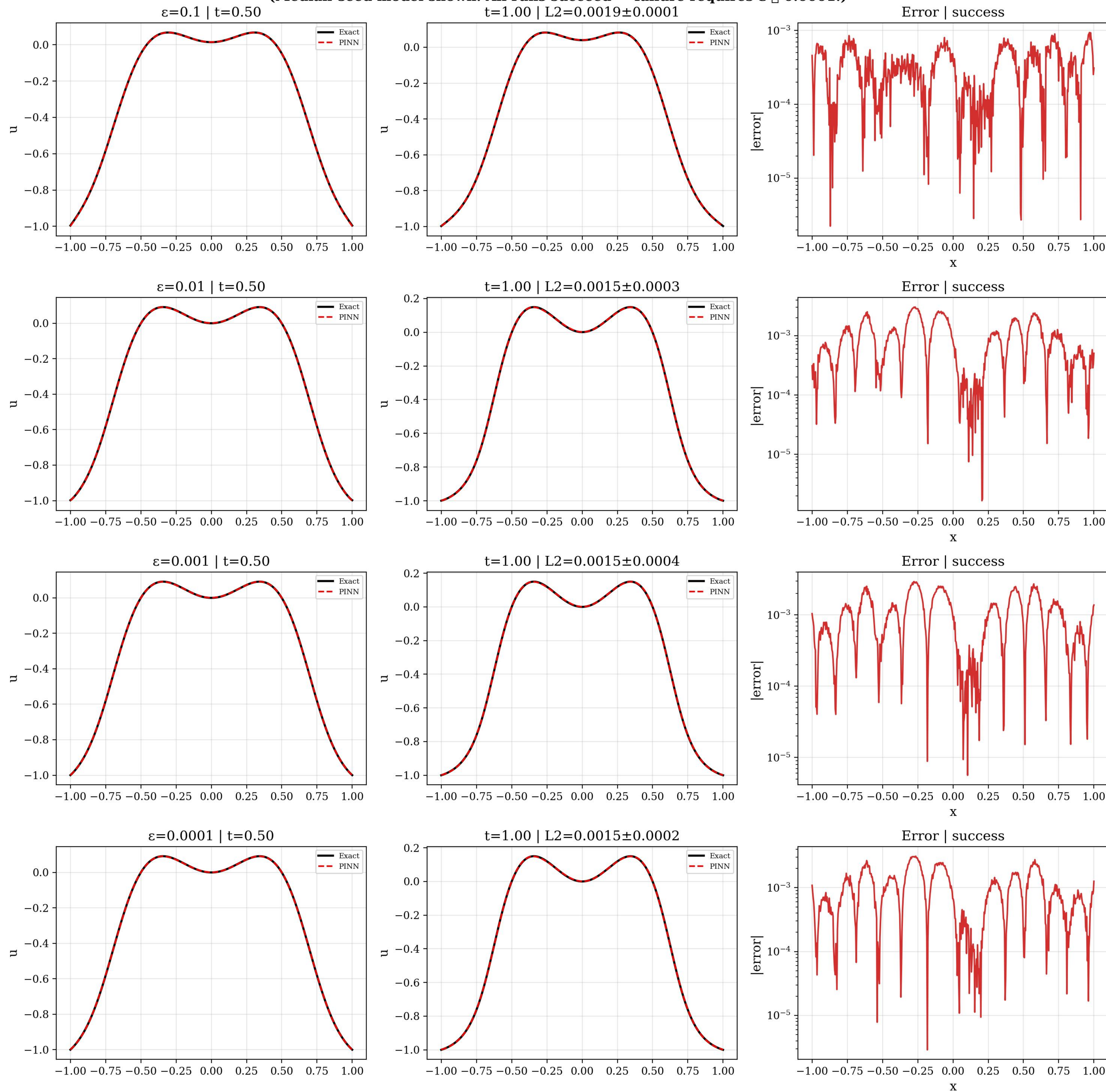


Allen-Cahn: PINN vs Exact for Different ε (Median-seed model shown. All runs succeed — failure requires $\varepsilon \leq 0.0001$.)



NOTE: All ε values succeed ($L2 < 0.1$). No visual failure is observable. PINN tracks exact solution in all cases. The interface/bulk ratio trend (not L2 failure) is the primary finding — see [interface_ratio_trend.png](#).